

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

H/w

(Gabbar N-S)

WORK, POWER, ENERGY

LECTURE-5

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision ✓
- 2 Gabbar Numerical Series
- 3 ..
- 4 ..
- 5 ..

“SMALL DAILY
IMPROVEMENTS LEAD
TO BIG RESULTS”

Recap From 1st Lecture

Work

- mode of transfer of energy

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$

Work

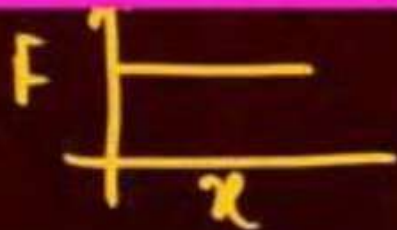
W = +ive

W = -ive

W = 0

Work

Work done by constant



$$W = \vec{F} \cdot \vec{x}$$

Work done by variable force



$$W = \int F \cdot dx$$

Energy

- Capacity to do work

$$m \cdot E$$

K.E

+

P.E

$$K.E = \frac{1}{2}mv^2$$

(Derivation)

✓
✓
✓
Detail
next
Lecture

Recap From 2nd Lecture

• Work-Energy Theorem

$$W = \Delta K \cdot E$$

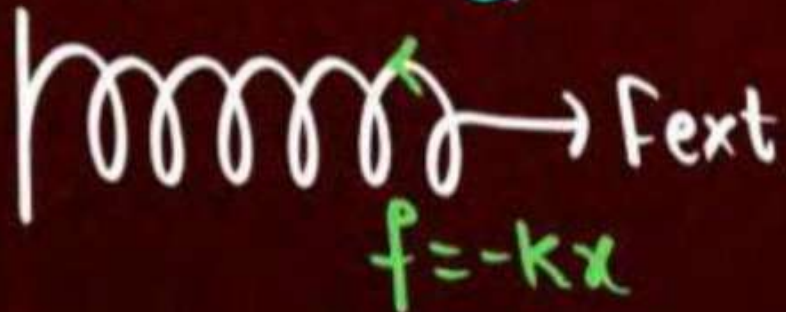
• P.E / U → Potential Energy

Zabardasti ✓

Relative position of particle

✓ Gravitation $P.E = mgh$

✓ Spring $P.E = \frac{1}{2} kx^2$



• WORK done by spring force

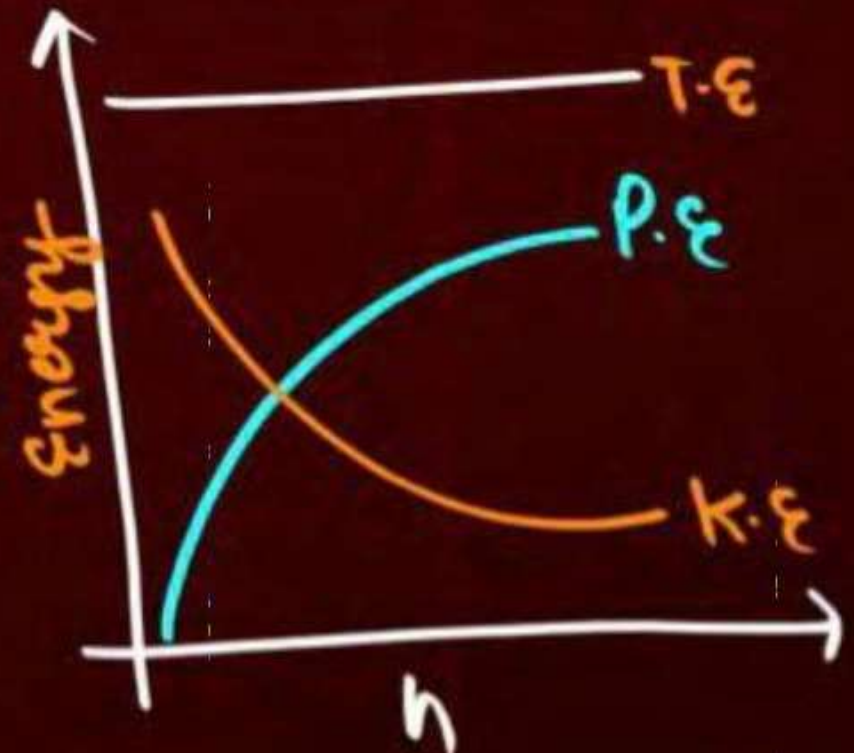
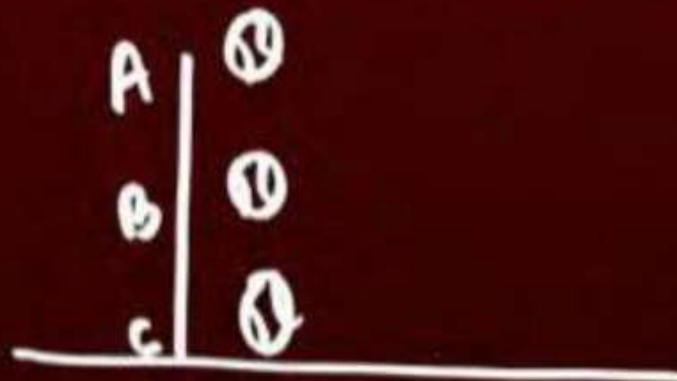
$$W = -\frac{1}{2} kx^2$$

✓ Law of Conservation of Energy

$$T.E_i = T.E_f$$

$$m.E_i = m.E_f$$

100J 100J



Recap From 3rd Lecture

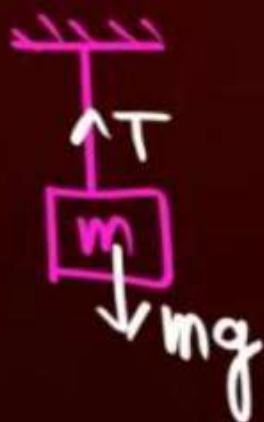
• Conservative Force

- Work $\downarrow$ depend on path.
 do not
- ✓ Gravitation Force ✓
- ✓ Spring / Electrostatic Force

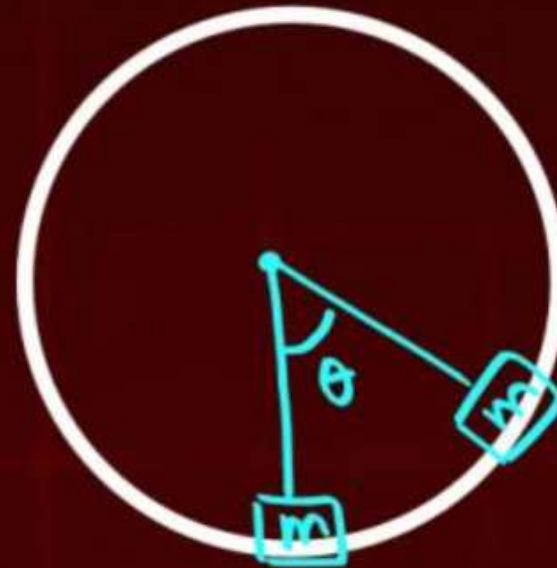
• Non-conservative force

- Work does depend on path
- ✓ Friction

Tension ✓



• Vertical Circular Motion



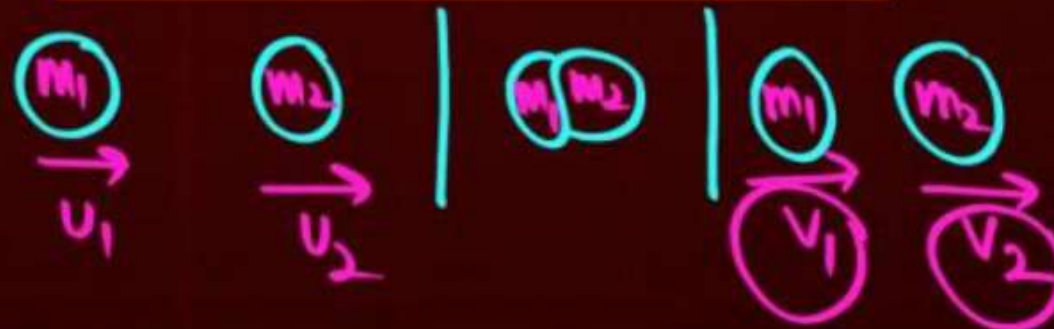
$$\checkmark \frac{mv^2}{R} = T - mg \cos \theta$$

$$\checkmark T = \frac{mv^2}{R} + mg \cos \theta$$

Collision

$P \rightarrow \text{cons.}$ $K.E \rightarrow \text{conserved}$

Perfectly elastic Collision

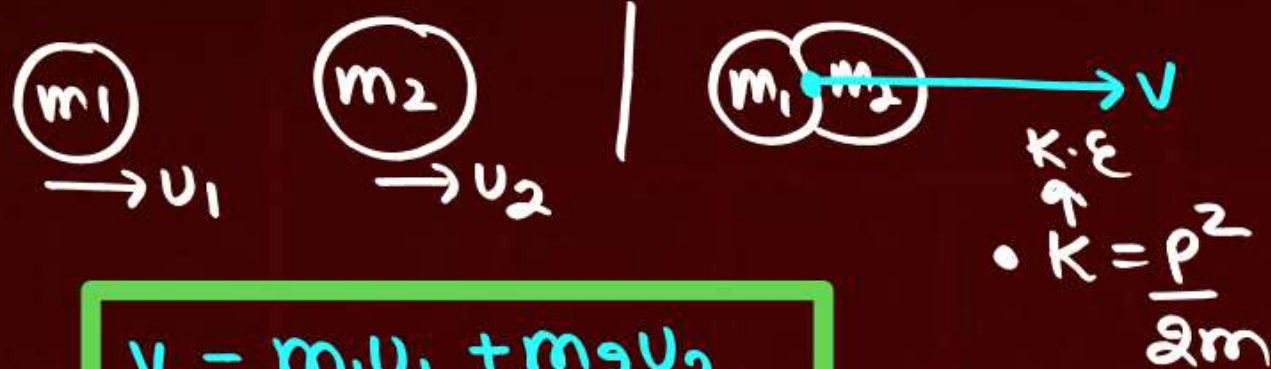


Perfectly inelastic collision

1 on 1

Recap From 4TH Lecture

Inelastic Motion



$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

Coefficient of Restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$e = 0$ Perfectly inelastic collision
 $e = 1$ Perfectly elastic collision

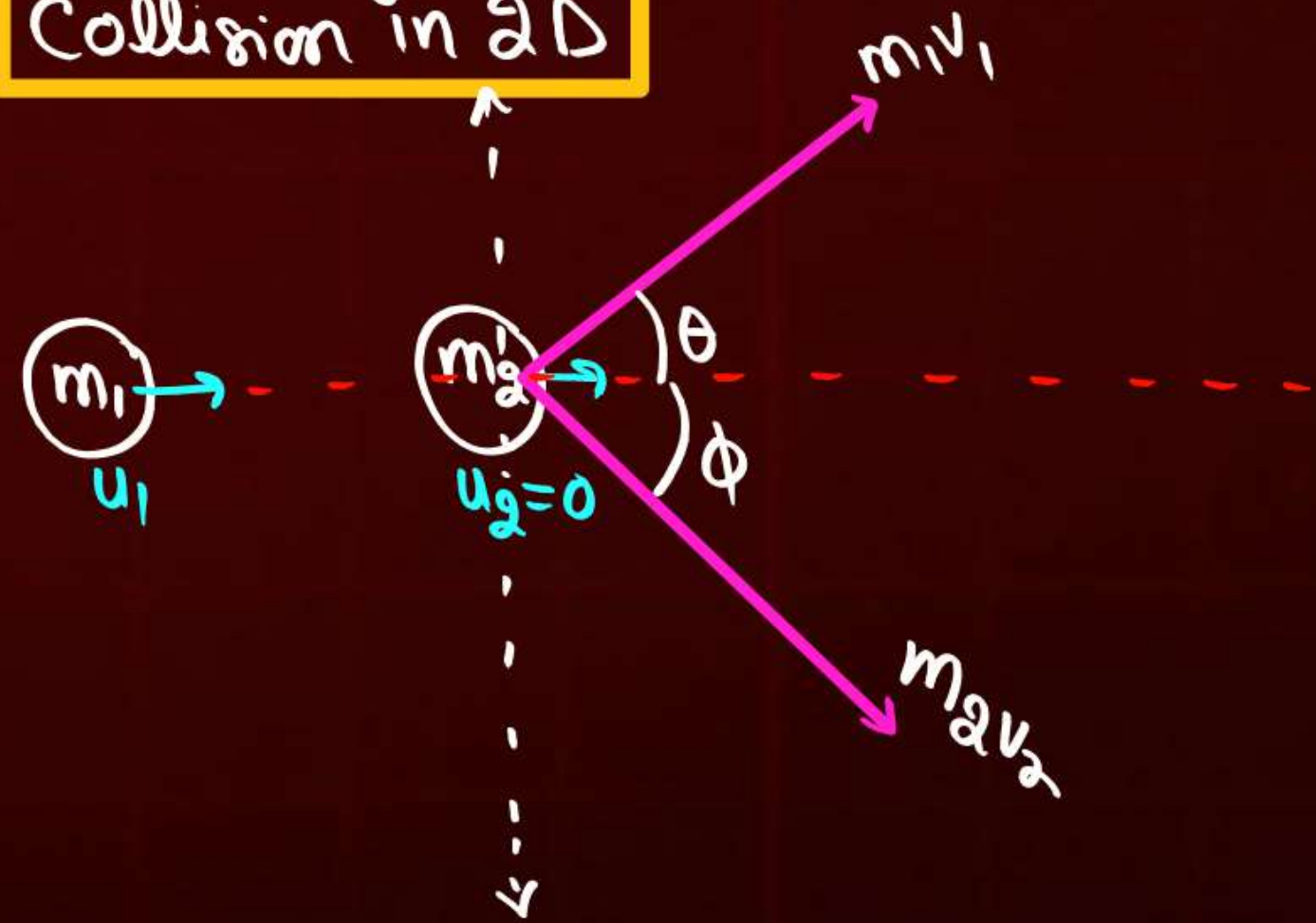
Power

$$P = \frac{W}{t} = F v \cos \theta = \frac{K \cdot \epsilon}{t} = \frac{P \cdot \epsilon}{t}$$

✓ watt
 ✓ Hp

$$1 \text{ Hp} = 746 \text{ watt}$$

Collision in 2D



Gabbar

→ Master
K
Saath
Kare ga -----

Numerical Series

PYQs

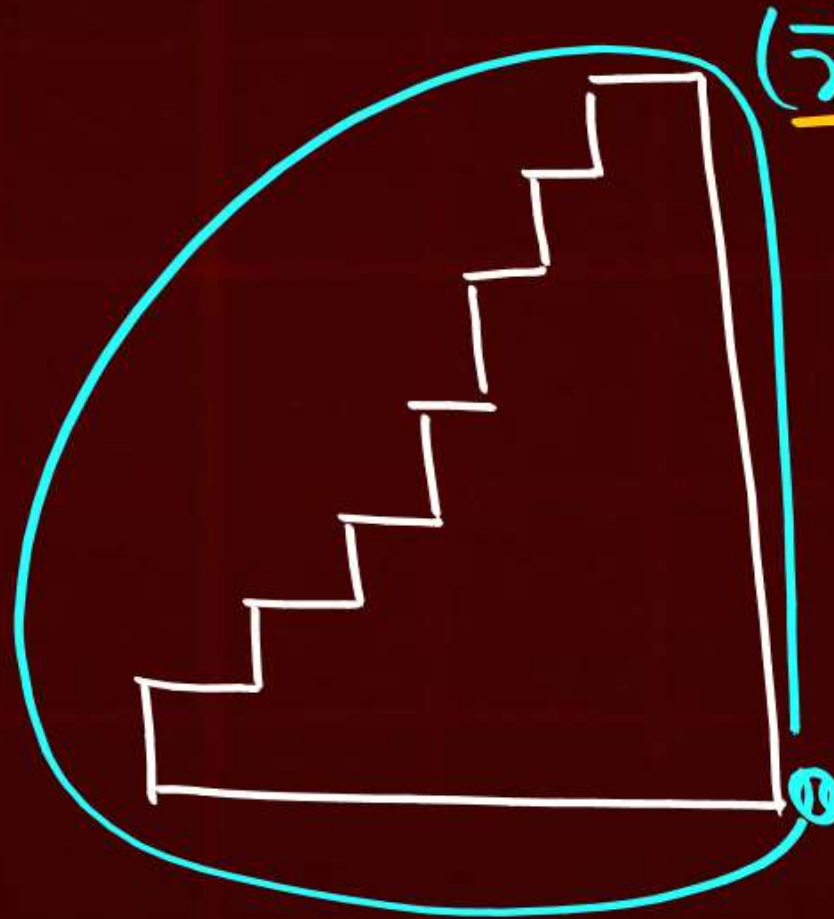
YE ASAAN HAI [SUMMER SERIES-A 2026]



④ The work done by a conservative force moving a particle around a closed path is:

Path^x

- A** Always positive
- B** Always negative
- C** Zero
- D** Depends on the path



($\vec{x}$) displacement = 0

$$W = \vec{F} \cdot \vec{x} = 0$$

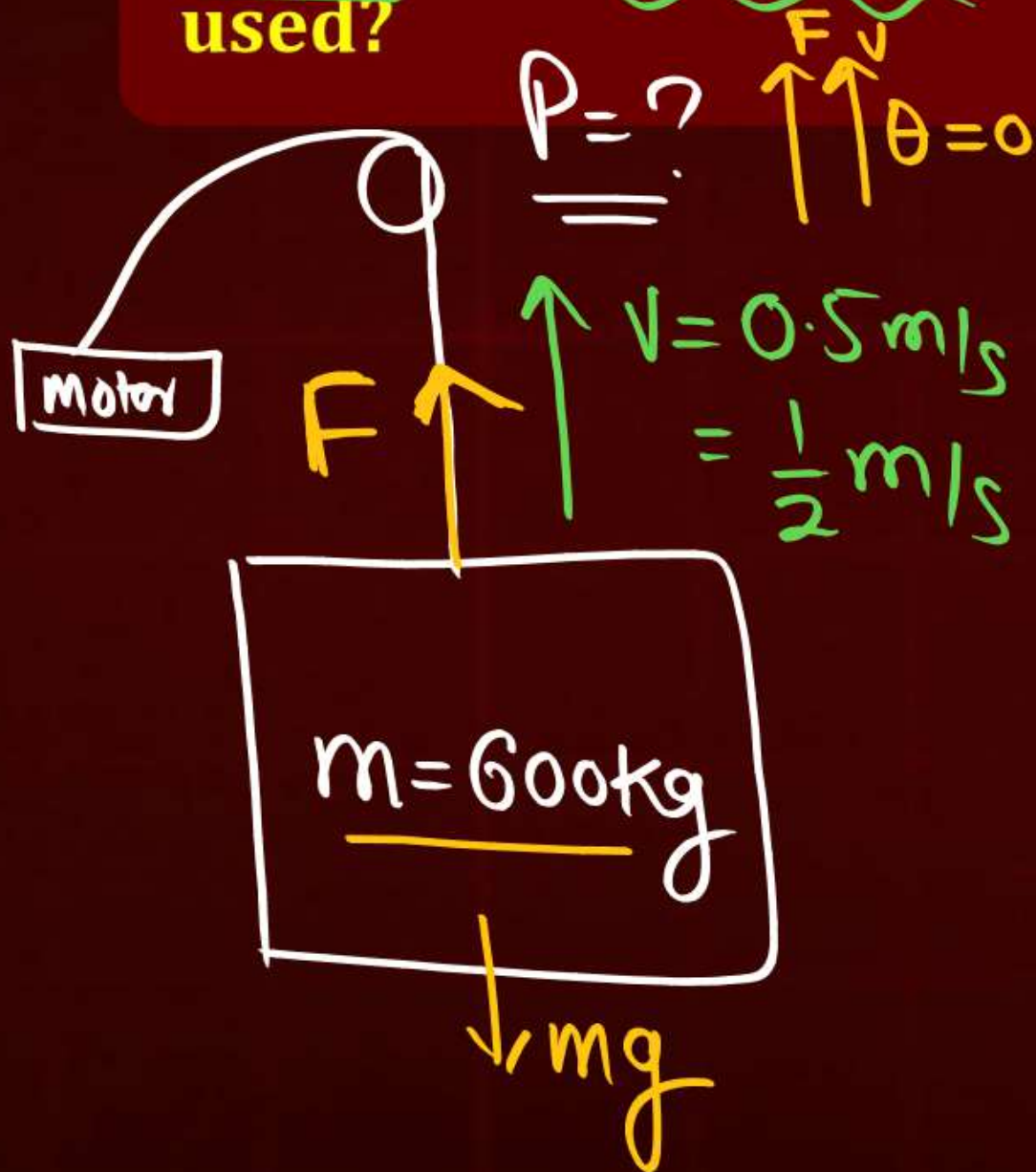
$$W = \vec{F} \cdot 0$$

$$\underline{W = 0 \text{ J}}$$

→ Hoojaye gear

TUM SE NA HO PAYEGA? [SUMMER SERIES-A 2026]

- ② An elevator weighing 600 kg is to be lifted up at a constant velocity of 0.5 m/s. What should be the minimum horse power of the motor to be used?



$$P = \vec{F} \cdot \vec{v}$$

$$P = Fv \cos \theta$$

$$P = Fv \cos 0$$

$$P = Fv$$

$$P = 6000 \times \frac{1}{2} = \frac{6000}{2} = 3000 \text{ watt}$$

$$P = 3000 \text{ watt}$$

$$P = 3000 \times 1 \text{ watt}$$

Hp

$$F = \text{weight}$$

$$F = mg$$

$$F = 600 \times 10$$

$$F = 6000 \text{ N}$$

$$g = 9.8 \text{ m/s}^2$$

$$g = 10 \text{ m/s}^2$$

$$1 \text{ hp} = 746 \text{ watt}$$

$$\frac{1}{746} \text{ hp} = 1 \text{ watt}$$

$$P = 3000 \times \frac{1}{746} \text{ hp}$$

$$\eta = 10$$

$$P = \frac{3000}{746} \text{ hp}$$

$$\checkmark P = 4 \text{ hp}$$

$$g = 9.8$$

$$\underline{P = 39 \text{ hp}}$$

YE EASY HAI? [SUMMER SERIES-A 2026]

What is meant by a collision? Discuss two types of collision.

2

YE EASY HAI? [SUMMER SERIES-B 2026]

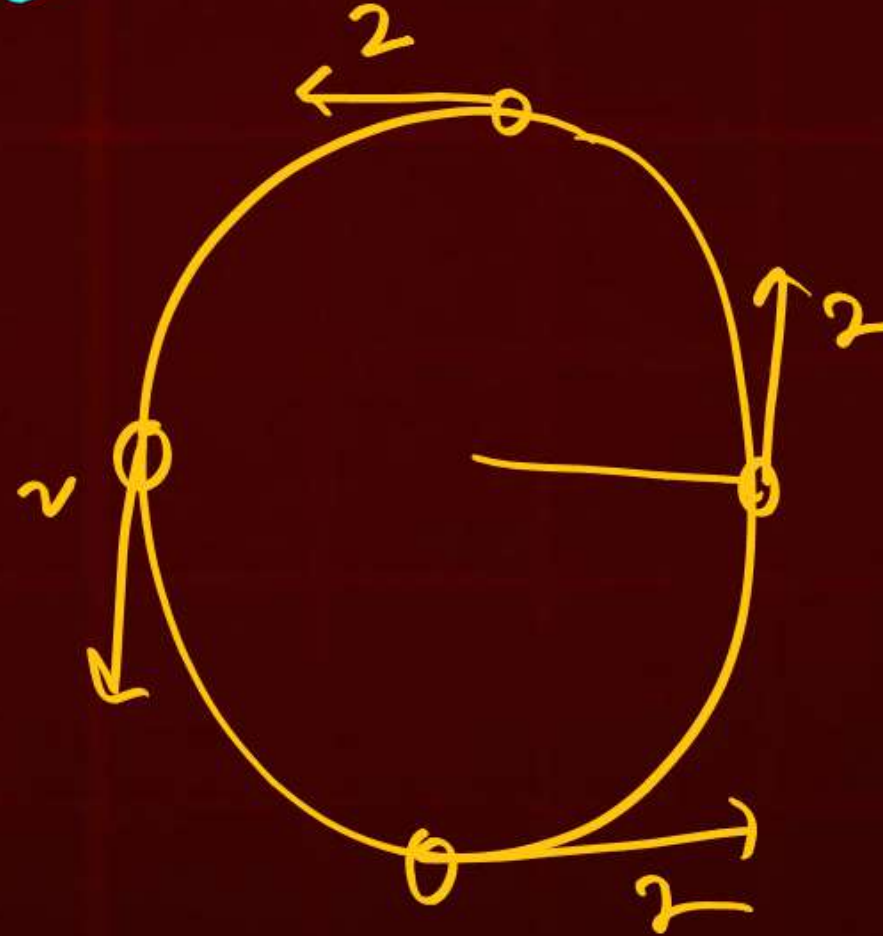
If a body's velocity changes direction but not magnitude, the motion is:

(NLM)

$V \rightarrow \text{change}$

$$\vec{V} = \underline{m \times \vec{D}}$$

- A** Rectilinear
- B** Uniform linear
- C** Uniform circular ✓
- D** Projectile



✓✓

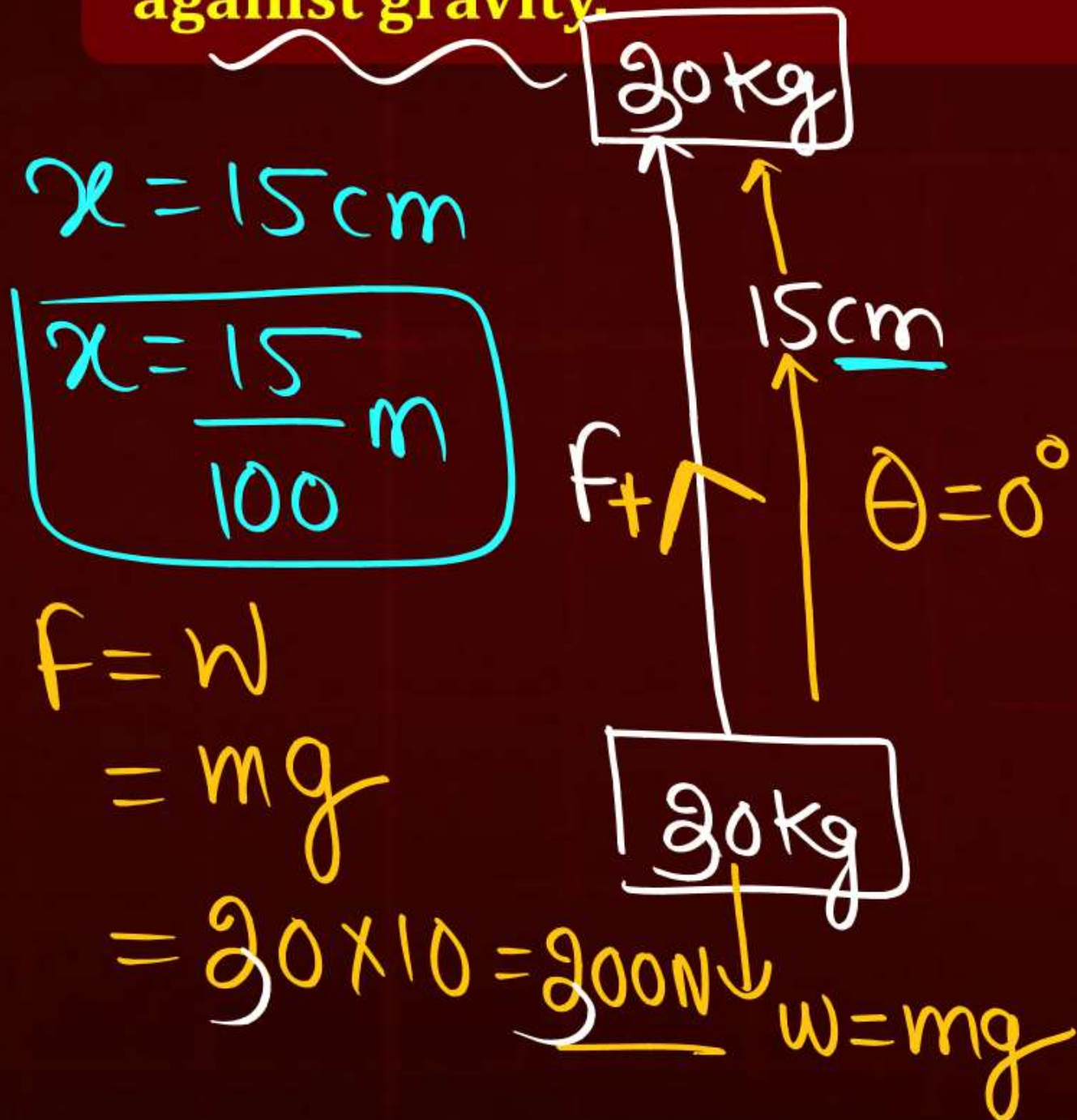
YE EASY HAI? [SUMMER SERIES-B 2026]

Which of the following forces is an example of conservative forces?

- A** Air resistance *✗*
- B** Viscous force *✗*
- C** Gravitational force *✓*
- D** None of these

TUM SE NA HO PAYEGA? [SUMMER SERIES-B 2026]

To lift a mass of 30 kg to a height of 15 cm, calculate the work done against gravity.



$$W = Fx \cos \theta$$

$$W = Fx \cos 0$$

$$W = Fx$$

$$= 300 \times \frac{15}{100}$$

$$= 45 \text{ J}$$

YE EASY HAI? [SUMMER SERIES-C 2026]

Which of the following is an example of a non-conservative force?

A Gravitational force ✓

B Spring force ✓

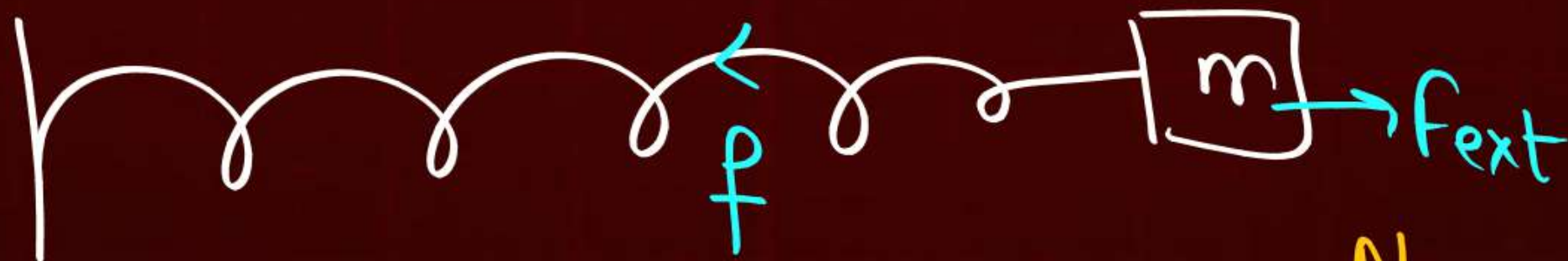
C Frictional force ✓

D Electrostatic force ✓

CAN YOU SOLVE THIS? [SUMMER SERIES-A 2025]

What is meant by restoring force and force constant?

Ans:- The Restoring force is the Spring force which try to bring the spring in its original position.



$$f \propto x$$

$$f = -kx$$

Spring
Constant

$$\frac{N}{m} = \frac{f}{x} = k$$

asam hai

Neet - 2017



DIFICULT HAI? QUESTION [SUMMER SERIES-A 2025]

A body of mass 10 kg initially at rest is subjected to a force of 40 N. What is the kinetic energy acquired by the body at the end of 10 sec?

$F = 40 \text{ N}$
 $m = 10 \text{ kg}$
 $u = 0$
 $a = 4 \text{ m/s}^2$
 $t = 10 \text{ sec}$
 $m = 10 \text{ kg}$
 $v = ?$

$F = ma$
 $\frac{F}{m} = a$
 $a = \frac{40}{10} = 4$

$v = u + at$
 $v = 0 + 4 \times 10$
 $v = 0 + 40$
 $v = 40 \text{ m/s}$

$\# K.E = \frac{1}{2}mv^2$
 $K.E = \frac{1}{2} \times 10 \times (40)^2$
 $K.E = \frac{1}{2} \times 10 \times 1600$
 $K.E = 8000 \text{ J}$

YE EASY HAI? [SUMMER SERIES-A 2025]

Distinguish between elastic and inelastic collisions. ✓

YE WALA ASAN HAI [SUMMER SERIES-C 2025]

The correct relationship between Calorie and Joule is

- A** 1 Calorie = 4.2 Joules ✓
- B** 1 Calorie = $1/4.2$ Joules
- C** 0.24 Calorie = 1 Joule ✓
- D** Both (A) and (C)

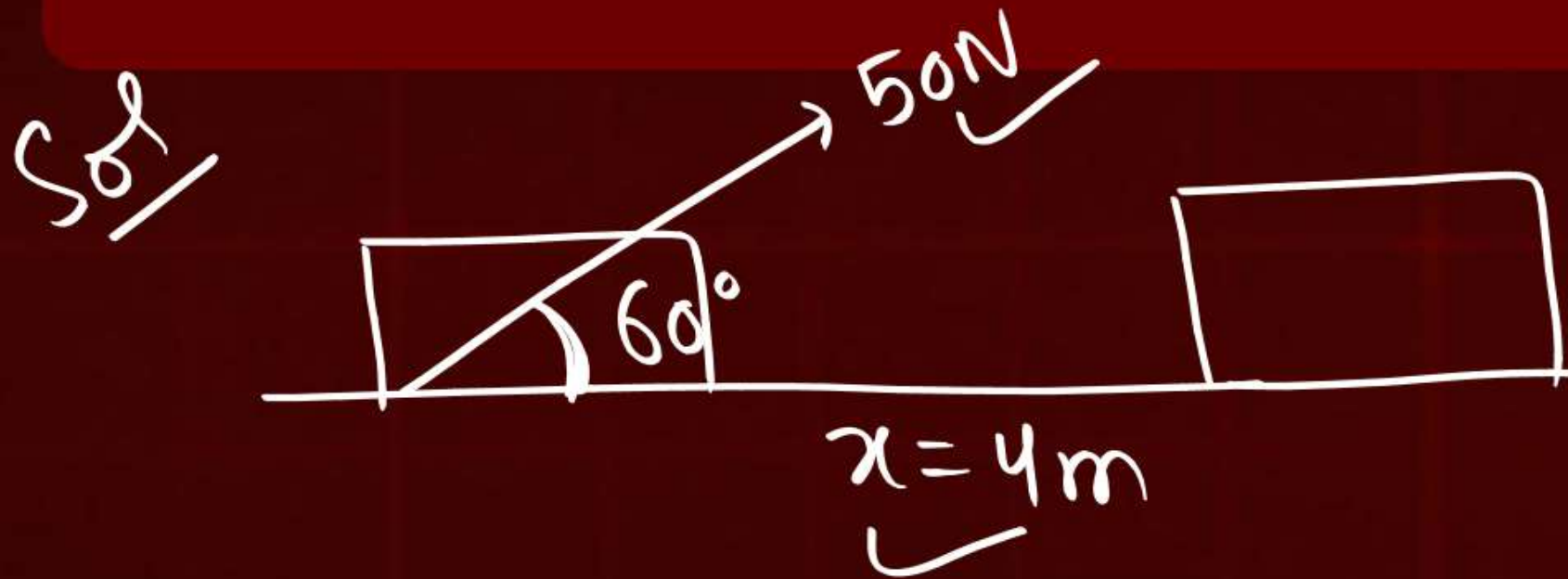
$$1 \text{ Cal} = 4.2 \text{ Joule}$$

$$\frac{1}{4.2} \text{ Cal} = 1 \text{ Joule}$$

$$0.24 \text{ Cal} = 1 \text{ Joule}$$

TENSAN HONE LAGA [JKBOSE JAMMU-2025-A]

A force of 50 N acts on a body and moves it to a distance of 4 m on a horizontal surface. Calculate work done if the direction of force is at an angle of 60° to the horizontal surface.



$$\begin{aligned}
 W &= F x \cos \theta \\
 &= 50 \times 4 \times \cos 60^\circ \\
 &= 200 \times \frac{1}{2} \\
 &= 100 \text{ J} \checkmark
 \end{aligned}$$

CAN YOU SOLVE THIS? [JKBOSE JAMMU-2025-A]

Give two points of difference between conservative and non-conservative forces.

YE WALA EASY HAI [JKBOSE JAMMU-2025-A]

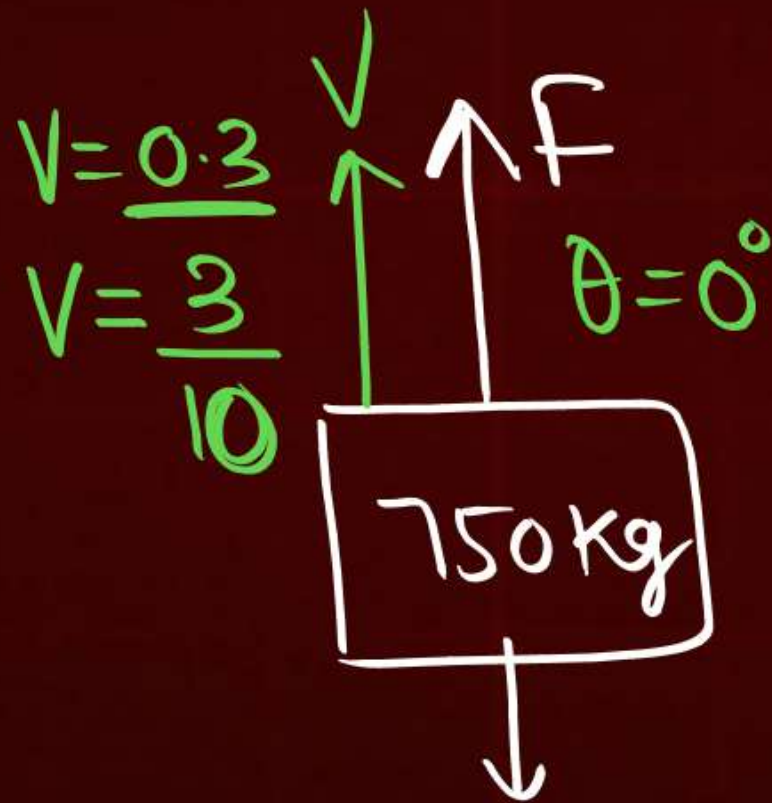
What is Collision? Define two types of collision.

A decorative yellow wavy line and a checkmark are positioned below the text.

TUMSE NHI HOGA? [JKBOSE JAMMU-2025-B]

Imp

An elevator weighing 750 kg is to be lifted up at constant velocity of 0.3 m/s. What should be the minimum horse power of the motor to be used?



$$F = W$$

$$= mg$$

$$= 750 \times 10$$

$$F = 7500 \text{ N}$$

$$P = FV \cos \theta$$

$$P = FV \cos 0^\circ$$

$$P = FV$$

$$P = 7500 \times \frac{3}{10}$$

$$P = 7500 \times 3$$

$$= 22500 \text{ watt}$$

$$P = \frac{22500}{746} \text{ Hp}$$

$$= 3 \text{ Hp} \checkmark$$

CAN YOU SOLVE THIS [JKBOSE JAMMU-2025-B]

Define power and energy. Give their units. ✓

YE TO ASAN HAI [JKBOSE JAMMU-2025-B]

What is collision? Define two types of collision.

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2025-C]



$$\uparrow f = -kx \uparrow$$

2019

2014
JKCET

✓ A force $F = (10 + 0.5x)$ acts on a particle in x direction, where F is in newton and x is in metre. Find the work done by this force during a displacement from $x = 0$ to $x = 2\text{m}$. (9th)

Sol ($F = 10 + 0.5x$)

✓ $F = 10 + \frac{1}{2}x$ (Variable Force)

$$W = \int_{x_1}^{x_2} F dx$$

$$W = \int_0^2 (10 + \frac{1}{2}x) dx$$

$$W = \int_0^2 (10 + \frac{1}{2}x) dx$$

$$W = \int_0^2 (10x^0 + \frac{1}{2}x^1) dx$$

$\int x^n dx = \frac{x^{n+1}}{n+1}$

$$W = \int_0^2 (10x^0 + \frac{1}{2}x^1) dx$$

~~SKB~~

$$W = \left[\underset{UL}{10} \times \overset{x}{2} + \frac{\overset{x^2}{2^2}}{4} \right] - \left[\underset{LL}{\overset{0}{10}} \times \overset{0}{0} + \frac{\overset{x^2}{0^2}}{\cancel{4}} \right]$$

$$W = \left[\frac{10x^{0+1}}{0+1} + \frac{1}{2} \frac{x^{1+1}}{(1+1)} \right]_0^2$$

$$W = \left[20 + \frac{1}{4} \right] - 0$$

$$W = 20 + 1$$

$$= 21 \text{ J}$$

$$W = \left[\frac{10x^1}{1} + \frac{1}{2} \frac{x^2}{2} \right]_0^2$$

$$W = \left[10x + \frac{x^2}{4} \right]_0^{\textcircled{2}}$$

YE EASY HAI? [JKBOSE JAMMU-2025-C]

How do potential energy of a spring vary with displacement?

$$\uparrow P.E = \frac{1}{2} k x^2 \uparrow$$

YE KR PAOGE? [JKBOSE JAMMU-2025-C]

What is Collision? Define two types of collision. ✓

YE TO ARAM SE HO JAYEGA [JKBOSE JAMMU 2024-X]

✓ A light and a heavy body have equal kinetic energy. Which has greater momentum?

- A** Heavy body ✓
- B** Light body ✗
- C** Both have equal momentum ✗
- D** Information is incomplete ✗

$$K.E = K.E$$

$$m \downarrow \quad m \uparrow \quad \checkmark \quad p \uparrow$$

$$p = \sqrt{2mK.E}$$

↑
greater

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-X]

Define Work and Power. Give their S.I. units. ✓

AAO SOLVE KRTE HAI [JKBOSE JAMMU 2024-X]

Q A force vec $\vec{F} = (2 + x)\text{N}$ acts on a particle in x-direction in metre. Find the work done by this force during a displacement from $x = 1\text{m}$ to $x = 2\text{m}$. $W = ?$

$F = (2 + x)$ Variable force $x_1 = 1\text{m}$ $x_2 = 2\text{m}$

Sol $W = \int_{x_1}^{x_2} F dx$

$W = \int_1^2 (2x^0 + x^1) dx$

$W = \int_1^2 (2 + x) dx$ $W = \left[\frac{2x^{0+1}}{0+1} + \frac{x^{1+1}}{1+1} \right]_1^2$

$W = \int_1^2 (2 + x) dx$ $W = \left[\frac{2x^1}{1} + \frac{x^2}{2} \right]_1^2$

$W = \left[2x + \frac{x^2}{2} \right]_1^2$

$W = \left[2 \times 2 + \frac{2^2}{2} \right] - \left[2 \times 1 + \frac{1^2}{2} \right]$

$W = \left[4 + \frac{4}{2} \right] - \left[2 + \frac{1}{2} \right]$

$$w = [4+2] - [2+\frac{1}{2}]$$

$$w = (6) - (2 + 0.5)$$

$$w = 6 - 2.5$$

$$w = 3.5 \text{ J} \checkmark$$

UE TO EASY HAI? [JKBOSE JAMMU 2024-Y]

A light and a heavy body have equal momentum. Which has greater kinetic energy?

- A** Heavy body
- B** ✓ Light body ($m \downarrow$) ✓
- C** Both have equal momentum
- D** Information is incomplete

$$p = p$$

$$m \downarrow \quad m \uparrow$$

$$\uparrow \quad K.E = \frac{p^2}{2m} \quad \downarrow$$

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-Y]

Derive work-energy theorem. ✓

YE MUSHKIL HAI KYA? [JKBOSE JAMMU 2024-Y]

H/W
Q

A particle moves with a velocity of $\vec{v} = (6\hat{i} - 4\hat{j} + 3\hat{k})\text{m/s}$ under the influence of a constant force:

$$\vec{F} = (20\hat{i} + 15\hat{j} - 5\hat{k})\text{N}$$

JEE-2016 ✓

Find the instantaneous power applied to the particle.

$$\vec{v} = 6\hat{i} - 4\hat{j} + 3\hat{k}$$

$$\vec{F} = 20\hat{i} + 15\hat{j} - 5\hat{k}$$

$$P = \vec{F} \cdot \vec{v}$$

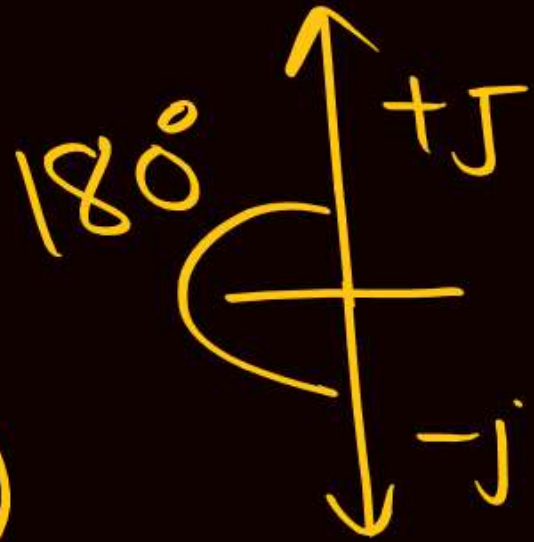
$$P = 120 - 60 - 15$$

$$P = 60 - 15$$

$$= \underline{45} \text{ watt} \checkmark$$

$$\vec{F} = 20\hat{i} + 15\hat{j} - 5\hat{k}$$

$$\vec{v} = 6\hat{i} - 4\hat{j} + 3\hat{k}$$



$$P = \vec{F} \cdot \vec{v}$$

$$= (20\hat{i} + 15\hat{j} - 5\hat{k}) \cdot (6\hat{i} - 4\hat{j} + 3\hat{k})$$

$$= \underbrace{20\hat{i} \cdot 6\hat{i}}_{\theta=0} + \underbrace{20\hat{i} \cdot -4\hat{j}}_{\theta=90^\circ} + \underbrace{20\hat{i} \cdot 3\hat{k}}_{\theta=90^\circ} + \underbrace{15\hat{j} \cdot 6\hat{i}}_{\theta=90^\circ} + \underbrace{15\hat{j} \cdot -4\hat{j}}_{\theta=180^\circ} + \underbrace{15\hat{j} \cdot 3\hat{k}}_{\theta=90^\circ} + \underbrace{-5\hat{k} \cdot 6\hat{i}}_{\theta=90^\circ} + \underbrace{-5\hat{k} \cdot -4\hat{j}}_{\theta=90^\circ} + \underbrace{-5\hat{k} \cdot 3\hat{k}}_{\theta=180^\circ}$$

$$= 20 \times 6 \cos 0^\circ + 20 \times -4 \cos 90^\circ + 20 \times 3 \cos 90^\circ + 15 \times 6 \cos 90^\circ + 15 \times -4 \cos 180^\circ + 15 \times 3 \cos 90^\circ - 5 \times 6 \cos 90^\circ + -5 \times -4 \cos 90^\circ + -5 \times 3 \cos 180^\circ$$

$$= 20 \times 6 \cos 0^\circ + 15 \times 4 \cos 180^\circ + 5 \times 3 \cos 180^\circ$$

$$P = \underline{20 \times 6} \cos 0^\circ + \underline{15 \times 4} \cos 180^\circ + 5 \times 3 \cos 180^\circ$$

$$P = 120 + 60 \times -1 + 15(-1)$$

$$= 120 + -60 + -15$$

$$= 120 - 60 - 15$$

$$= 60 - 15$$

$$= \underline{45} \text{ watt } \checkmark$$

CAN YOU SOLVE THIS? [JKBOSE JAMMU 2024-Z]

Define coefficient of restitution. How does it define elastic and inelastic collision?

$$e=1$$

$$e=0$$

YE MUSHKIL HAI? [JKBOSE JAMMU 2024-Z]

A force $\vec{F} = (2 + x)\text{N}$ acts on a particle in x -direction in metre. Find the work done by this force during a displacement from $x = 1\text{m}$ to $x = 2\text{m}$.

$$W = 3.5\text{J}$$

SUMMARY



✓
Gabber numerical
Session

HOMework

H/w

$$\vec{F} = 6\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\vec{v} = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$P = ?$$

$$P = \vec{F} \cdot \vec{v}$$

Watt

H/w

$$\vec{F} = 6\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\vec{x} = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$W = ? \quad W = \vec{F} \cdot \vec{x}$$

Joule



SMALL DAILY
IMPROVEMENTS LEAD
TO BIG RESULTS

Thank
You

